

Chapter 1

General background

Humans are multicellular organisms composed of hundreds of cell types. Every cell in an organism has deoxyribonucleic acid (DNA) stored in the cell's nucleus. DNA is the hereditary material that contains the instructions for a cell to perform its functions, divide, and differentiate. The cells in an organism must also communicate with each other and the environment to coordinate all the processes. Four bases, adenine (A), thymine (T), cytosine (C), and guanine (G), are composing DNA sequences (genetic code) to encode a blueprint of life. All genetic material within a single cell is known as a genome. Specific sequences within DNA, termed genes, are considered functional units of heredity. A subset of genes encode proteins (protein-coding), while other genes do not (non-coding). Proteins are molecules that perform most of the cellular functions. A gene is first copied (transcribed) into ribonucleic acid (RNA) during transcription. In the case of protein-coding genes, RNA molecules are then used in translation to inform what amino acids should be brought and connected into a chain to form a protein. RNA molecules transcribed from non-coding genes are further processed and participate in other cellular processes, such as transcription regulation.

Every cell in an organism has the same set of genes. However, cells are different and form different tissues and organs. What enables cells to differentiate into many types when the instructions are the same? Gene transcription is highly regulated, meaning specific cells express genes and their combinations when and where needed. Major players in this process are a particular class of proteins — transcription factors (TFs). They act by binding to specific DNA sequences (regulatory regions), which can control the associated gene transcription rate. Interactions between TFs and regulatory DNA sequences ensure gene transcription control in space and time. This chapter overviews basic molecular biology concepts, such as DNA, RNA, genome, and the process of transcription, to provide the background to dive deeper into how transcription is regulated.

1.1 Organization of the genome

In eukaryotes, the cell nucleus stores hereditary information as DNA. Cells copy (replicate) genomic DNA during division or transcribe portions of the DNA (genes) into RNA to produce proteins. Human DNA is only 2 nanometers wide, but if

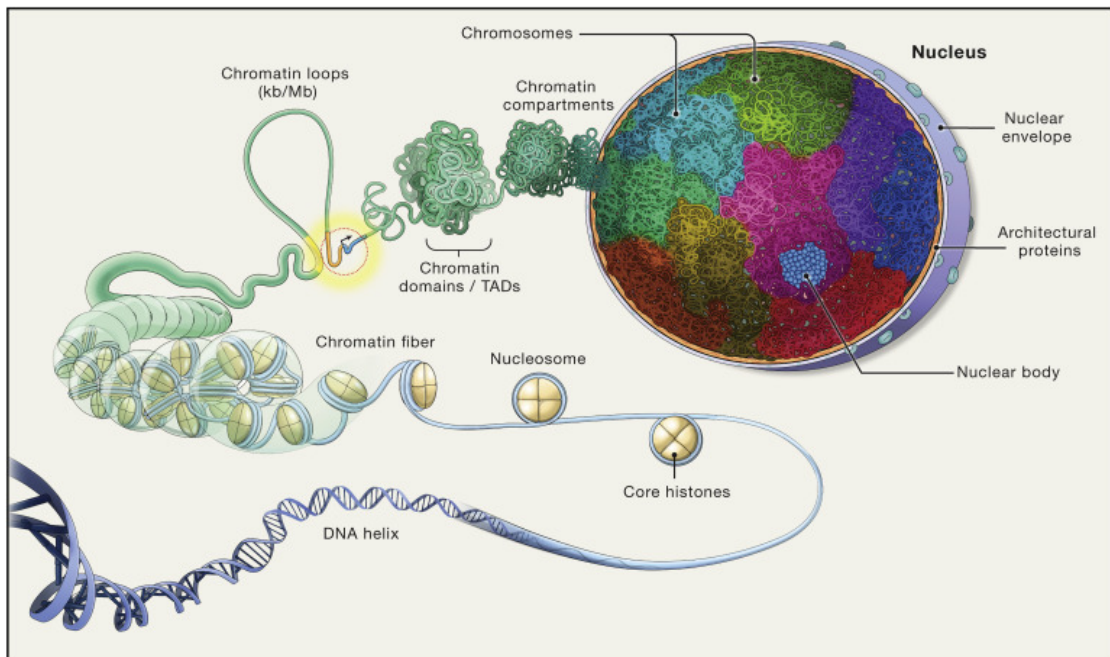


Figure 1.1: Eukaryotic genome organization. Genomic DNA is wrapped around histones to form nucleosomes, the basic units of chromatin fiber. Chromatin can form loops that span kilobase pairs (kb) or megabase pairs (Mb) and are essential for gene expression regulation. Bigger stretches of chromatin are organized into chromatin domains or topologically associated domains (TADs), forming chromatin compartments to pack everything into the nucleus. Each chromosome occupies a territory in the nucleus. The nucleus also contains a nuclear body and architectural proteins. A nuclear envelope surrounds the contents of the nucleus. Adapted from [2].

we stretch it into a single thread, it would be around 2 meters long [1]. The microscopic cell nucleus stores tightly compacted DNA. Multiple levels of the organization are applied. The lowest level of organization is a DNA helix wrapped around a specific class of proteins called histones. The highest level of organization is a chromosome, visible during the cell division. A chromosome consists of a single strand of DNA and various proteins. The human genome is organized into 22 pairs of numbered chromosomes and one pair of sex chromosomes (XX or XY). During interphase, DNA organizes into chromosome territories in the nucleus (Figure 1.1) [2]. In addition to the DNA and architectural proteins, the nucleus contains different types of nuclear bodies (nucleolus, paraspeckles), made of proteins and involved in cellular processes, such as telomere maintenance or stress response [2, 3]. A nuclear envelope surrounds the content of the nucleus. The specific organization ensures proper genome functioning — expressing genes when and where needed.

1.1.1 DNA

The existence of DNA was discovered in 1869 by Friedrich Miescher when he found a new substance inside the nuclei of the human white blood cells that did not have the same properties as proteins or lipids [4, 5]. In 1953, contributions of Rosalind

Franklin, James Watson, and Francis Crick led to the publication of resolved DNA molecule structure and is often considered the beginning of molecular biology [6]. This fundamental moment in science history allowed us to explain the possible mechanism of DNA replication, thus information multiplication and transfer.

DNA stores genetic information in the cell. In the DNA structure, the sugar backbone (deoxyribose) holds four main types of nucleotides: purines adenine (A) and guanine (G), and pyrimidines thymine (T) and cytosine (C) [7] [1]. Chargaff's rule states that in the DNA of any species, the proportions of A and T nucleotides and C and G are always equal, demonstrating unique base-pairing (complementarity), when forming a double-stranded DNA helix [7, 8]. A-T and C-G nucleotide pairs in two strands of the DNA and are held by two and three hydrogen bonds, respectively (Figure 1.2).

The most commonly found DNA structure is called a B form of DNA. Two helical DNA strands entwine to form spaces between the strands, major and minor grooves, which are 22 and 12 Å wide, respectively. The grooves create space for the proteins to interact with the double-stranded DNA. One of the most important classes of proteins that bind DNA is transcription factors (TFs) [9]. TFs are essential for the regulation of transcription. They act by recognizing ~~the~~ specific sequences of nucleotides and forming TF-DNA complexes.

1.1.2 Chromatin

Chromatin is a mixture of DNA and proteins that form chromosomes, the highest level of genome organization. The fundamental element of chromatin is a nucleosome, which consists of 146 bp of DNA wrapped around the core histone protein octamer. Nucleosomes gather into a chromatin fiber typically 5-24 nanometers in diameter. The fibers organize into higher structures. They fold into chromatin domains, called topologically associated domains (TADs) [10]. TADs associate with each other and create chromatin compartments. When cells are dividing, chromatin assembles into a metaphase chromosome. During interphase, chromatin is organized into domains and compartments and tends to localize in specific territories (Figure 1.1) [2, 11]. Chromatin fibers from different chromosomes interact at the edges but do not entangle. In addition to chromatin, the nucleus contains protein structures called nuclear bodies. Nuclear bodies, such as nucleolus, Cajal bodies, paraspeckles, and others, are membrane-less organelles [12]. Their functions include the coordination and processing of RNAs and coordination of chromatin-modifying proteins, among others [12]. The nuclear envelope surrounds the contents of the nucleus.

Canonical core histone proteins are H2A, H2B, H3, and H4, with two copies each in a single octameric histone core. A linker histone H1 separates every core nucleosome [11, 13]. The variants in specific genomic regions replace well-conserved canonical histone proteins. For example, H3.3 replaces the H3 histone in active genes and promoter regions [13, 14]. Histone protein N-terminal tails undergo extensive post-translational modifications (PTMs), such as acetylation and methylation, on their amino acids. These modifications are essential for controlling the accessibility of the chromatin. Accessible chromatin is accessed



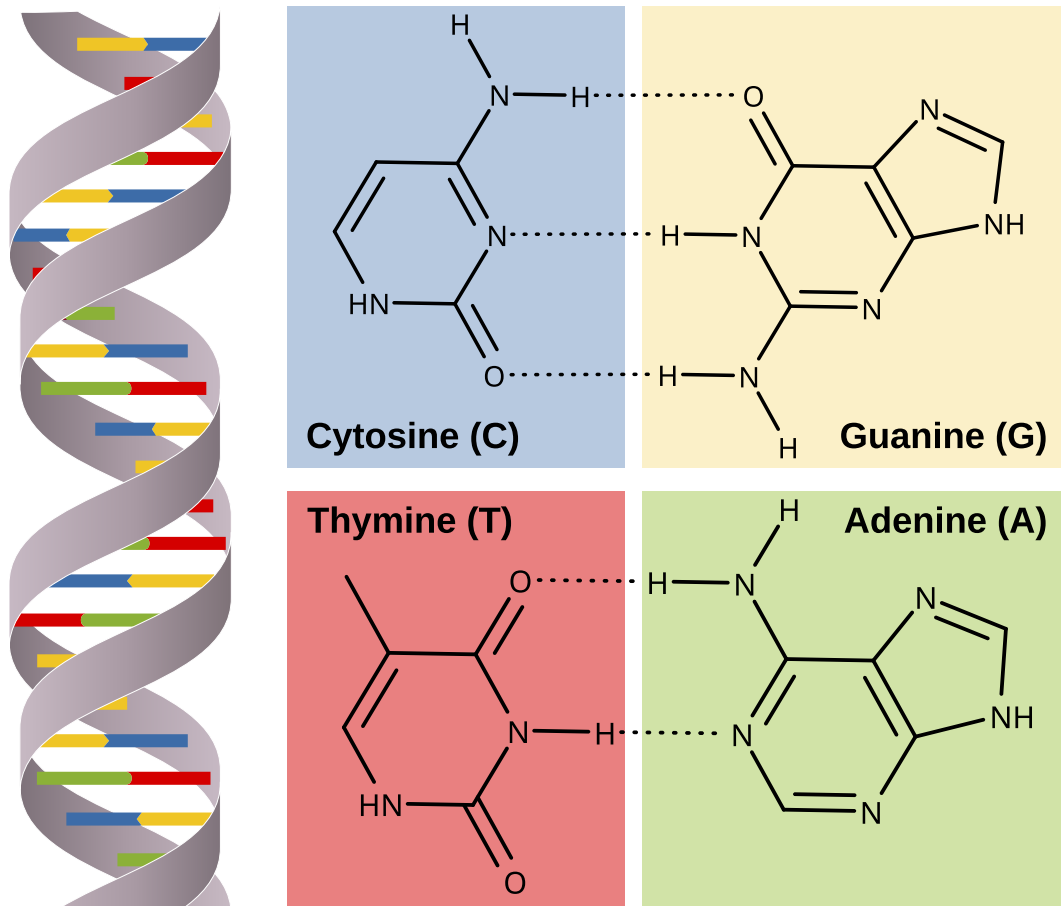


Figure 1.2: DNA structure. The two chains of the DNA are composed of a sugar backbone and four nucleotides: adenine (A), thymine (T), cytosine (C), and guanine(G). Nucleotides in the two chains connect via hydrogen bonds in a strict complementarity, where A is bound to T and C is bound to G.

by a transcriptional machinery and regulatory proteins [13].

Based on its activity state, chromatin can be heterochromatin or euchromatin [15]. Euchromatin is a more open, active chromatin with more widely spaced nucleosomes. Several histone modifications (Histone 3 Lysine 4 trimethylation (H3K4me3), Histone 3 Lysine 27 acetylation (H3K27ac), Histone 3 Lysine 36 methylation (H3K36me)) and histone variants (H3.3, H2A.Z, H2A.B) are marks of active transcription. Heterochromatin is generally known as a silent chromatin and is less accessible and transcribed. Constitutive heterochromatin is more static and contains Histone 3 Lysine 9 trimethylation (H3K9me3). Histone 3 Lysine 27 trimethylation (H3K27me3) marks the facultative heterochromatin, which can switch between active and inactive states. For example, facultative heterochromatin marks an inactive X-chromosome [11].

1.2 Gene transcription

Gene is a basic unit of heredity. It is a stretch of DNA in the genome and is subjected to transcription by RNA polymerase. Transcription is a process of copying DNA and producing an RNA molecule. Genes can be protein-coding or non-coding. The protein-coding RNA template is translated into a protein, while non-coding genes produce non-coding RNA molecules. A collection of proteins assembles into transcriptional machinery and organizes the transcription process.

1.2.1 RNA

Transcription machinery, with the leading player RNA polymerase, transcribes a portion of the genome and produces RNA. RNA molecule is single-stranded and consists of the sugar backbone (ribose) and four types of nucleotides: uracil (U), A, C and G (Figure 1.3). RNA molecules can adopt various secondary structures by forming hydrogen bonds between nucleotides (C pairs with G, while A pairs with U) within the same molecule [16].

The main types of RNAs, based on their functions, are messenger RNA (mRNA), ribosomal RNA (rRNA), transfer RNA (tRNA), and non-coding RNAs (ncRNAs) [16–18]. A precursor mRNA (pre-mRNA) is produced in the nucleus during transcription by copying a protein-coding gene (Figure 1.4) [17]. RNA polymerase synthesizes RNA from nucleotide triphosphates (NTPs) (ATP, GTP, CTP, and UTP, precursors of A, G, C, and U, respectively). A pre-mRNA is heavily processed to produce an mRNA that carries information to the cytoplasm, where ribosomes translate it to a polypeptide. mRNA has an open reading frame (ORF), consisted of introns and exons. The ORF is flanked by three prime untranslated region (3'UTR) and five prime untranslated region (5'UTR) [16, 19]. During splicing, a specific protein complex, spliceosome, processes most of the pre-mRNA by cutting out intronic (non-protein coding) gene segments to leave only exons (protein-coding) [20]. Introns can be spliced differently by choosing to leave out some exons, creating a variety of transcripts copied from the same gene. This process is known as alternative splicing [21]. Splicing factor proteins recognize splicing signaling sequences at the intron-exon boundaries [20, 21]. In addition to

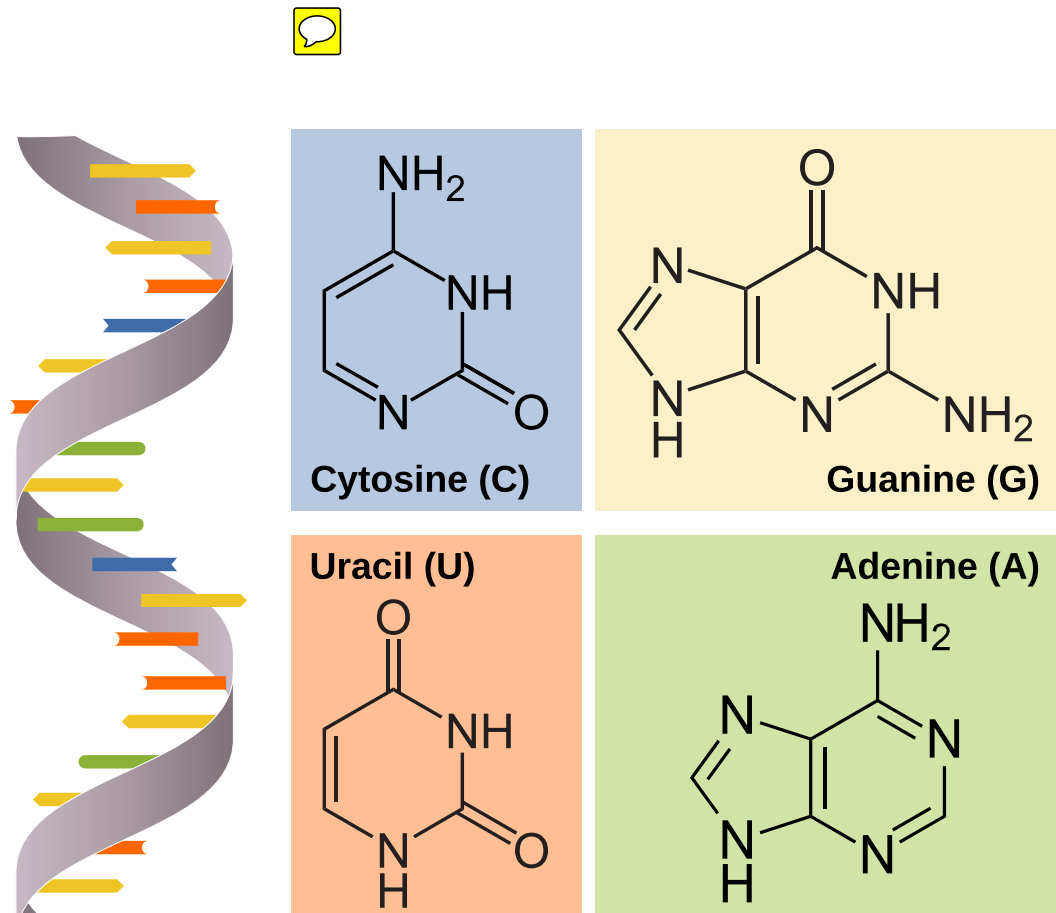


Figure 1.3: RNA structure. RNA is a single-stranded molecule composed of a sugar backbone and four nucleotides: uracil (U), adenine (A), cytosine (C), and guanine (G).

~~the~~ splicing, a cap is added at the 5' end of the RNA, while multiple added A nucleotides at the 3' end form a poly-A tail [21]. After this, mRNA **matures** and travels from the nucleus to the ribosomes in the cytoplasm. Ribosomes receive individual mRNAs and read the information encoded. Ribosomes contain rRNA, and each amino acid is brought into the ribosome by tRNAs that recognize every three mRNA nucleotides, termed codons [16, 18]. Codons code for amino acids.

Successive Longer than 200 nucleotides non-coding RNAs (long non-coding RNAs (lncRNAs)) and shorter small ncRNAs are the two main size-based categories of ncRNAs. Small ncRNAs can be divided into micro RNAs (miRNAs), small nucleolar RNAs (snoRNAs), small nuclear RNAs (snRNAs), small-interfering RNAs (siRNAs) and PIWI-interacting RNAs (piRNAs) [22]. miRNAs have an essential function in eukaryotic gene regulation. They can target and silence mRNAs to prevent translating them into proteins [17]. Aberrations of the ncRNA activities can be associated with diseases such as cancer. For example, miRNA can act as a tumor suppressor and target genes responsible for cell growth. snRNAs and snoRNAs are involved in splicing. snoRNAs can also guide the modifications of rRNAs [22]. The primary function of siRNAs is silencing the genomic locus, for example, the same locus from which siRNA is derived. piRNAs play a role in silencing the transposable elements [23].

1.2.2 RNA polymerase and transcriptional machinery

In eukaryotes, RNA polymerase II (Pol II) is responsible for transcribing genes encoding mRNAs and some ncRNAs [24]. This twelve-subunit protein uses NTPs as a substrate to produce ~~pre-m~~mRNA molecules from a DNA template. The RNA sequence synthesis process follows nucleotide complementarity. In the DNA template, nucleotides **A, C, G, and T** will be transcribed into RNA nucleotides **U, G, C, and A**, respectively.

Pol II is not able to recognize the transcription start sites (TSSs) by itself and needs to be guided by a set of proteins known as general transcription factors (GTFs) (Figure 1.4). Pol II and GTFs form a minimal transcription machinery, and all these proteins, together with the template DNA, form the preinitiation complex (PIC) [24, 25]. The GTFs are TFIIB, TFIID, TFIIIE, TFIIF, and TFIIH, and each plays a specific role in transcription initiation. They assemble one after another. Proteins TATA-binding protein (TBP) and TBP-associated factors (TAFs) compose TFIID and recognize a particular sequence of DNA near the TSS, called a TATA box. TFIIB and the non-phosphorylated form of Pol II join the complex, followed by TFIIF, TFIIIE, and TFIIH. Finally, Pol II is phosphorylated to proceed to elongation [24]. The transcriptional machinery produces a central portion of ~~m~~mRNA during transcription elongation. Transcription ends with the transcription termination process, and ~~pre-m~~mRNA travels for further processing and splicing.

Different cell types at constant rates express a limited set of genes called housekeeping genes. However, for cells to differentiate and perform cell-type specific functions, specific genes need to be transcribed at a particular time and place. Basic transcriptional machinery cannot decide what genes to express, but



isn't 5' UTR the 1st exon?

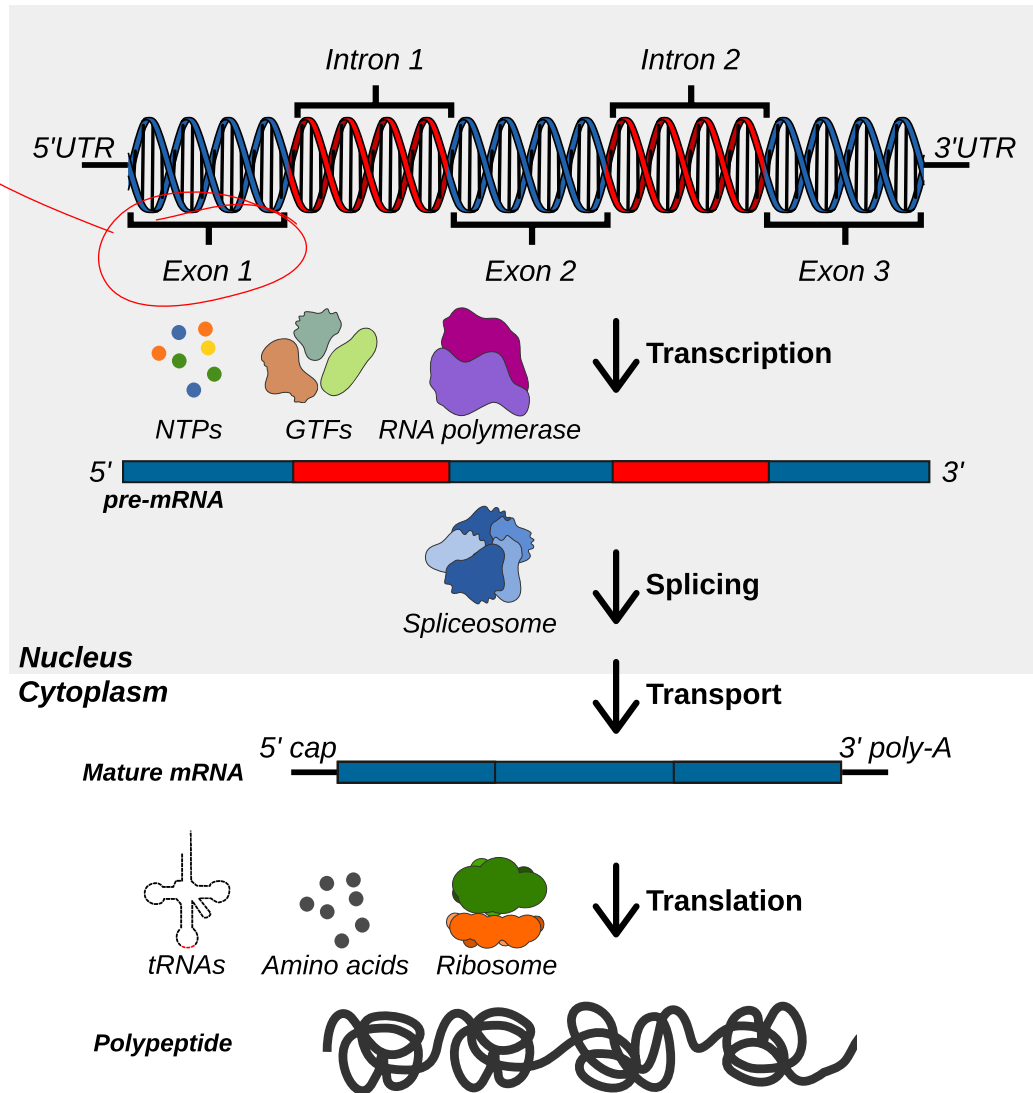


Figure 1.4: From gene to protein. During transcription, an open reading frame (ORF), flanked by 5' and 3' untranslated regions (UTRs), is transcribed by RNA polymerase to produce pre-mRNA. General transcription factors (GTFs) are part of the transcription complex, and RNA polymerase uses nucleotide triphosphates to produce the RNA molecule. Pre-mRNA has introns and exons. The spliceosome cuts out non-protein coding introns during splicing, leaving only protein-coding exons. 5' cap and 3' poly-A tail are added to produce a mature mRNA. In the cytoplasm, the ribosome translates mRNA into a polypeptide chain. tRNAs bring each amino acid into the ribosomes.

various regulatory processes make those instructions. For example, TFs bind DNA that forms regulatory regions. These regions are called promoters (proximal to the TSS) and enhancers (distal ~~upstream and downstream~~ from the TSS). TF-DNA interactions signal the PIC assembly at the associated gene promoter. Mechanisms of TF binding to the DNA and transcriptional regulation are very complex, and the field still lacks details on the organization of these processes. The next chapter overviews the regulatory regions, TFs, and transcription regulation.



Chapter 4

Summary of papers

This chapter summarizes the three publications that are the foundation of this thesis. The unifying **subject** of this thesis is the investigation of transcription factor-DNA interactions. I explored this **subject** through two main aims:

- (1) discover new TF binding profiles and update the JASPAR database while also developing and updating robust pipelines for automated data processing and post-processing;
- (2) identify and characterize co-binding DNA patterns.

4.1 Paper I. The release of the 2022 JASPAR database

Castro-Mondragon JA*, Riudavets-Puig R*, **Rauluseviciute I***, Lemma RB, Turchi L, Blanc-Mathieu R, Lucas J, Boddie P, Khan A, Manosalva Pérez N, Fornes O, Leung TY, Aguirre A, Hammal F, Schmelter D, Baranasic D, Ballester B, Sandelin A, Lenhard B, Vandepoele K, Wasserman WW, Parcy F, Mathelier A. *JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles*. Nucleic Acids Research. 2022 Jan 7;50(D1):D165-73; doi: 10.1093/nar/gkab1113.

Paper I describes the 2022 release of the JASPAR database. JASPAR is a widely used open-access repository of transcription factor binding motifs ~~created in and~~ maintained since 2004. The database is updated regularly with new motifs, features, and tools. One of the biggest strengths of JASPAR is that a team of experts manually curates motifs before they enter the database. We manually curate two types of motifs: (1) in-house *de novo* discovered motifs from **peak** data and (2) external motifs from individual papers or databases. We mine **peak** data from databases (for example, ReMap [129], and GTRD [130]) or from publications. Currently, JASPAR has two motif collections: CORE and UNVALIDATED. The CORE collection contains profiles with orthogonal support. The UNVALIDATED collection has profiles of the same sound quality but without validation by independent orthogonal experiments.

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individual

In 2022, the CORE and UNVALIDATED collections were complemented with 501 and 182 profiles, respectively. The CORE collection increased by 19% compared to the previous release. In addition to expanding the motif collections, we revisited the existing motifs. We revised the UNVALIDATED collection profiles to check for possible new validations that would move the profiles into the CORE collection. In 2022, we upgraded 52 such profiles. We extensively revised the metadata inconsistencies. Metadata fields, such as TF class and family, were updated with the newest available information.

the Together with the motif collections, we provide additional features such as motif clustering, familial TF binding motifs, TFBS predictions, and genome browser tracks. We individually clustered CORE collection motifs for each taxon and CORE and UNVALIDATED collection motifs. Clusters are visualized as radial trees and annotated with TF class information. Furthermore, we used the motif clusters to construct familial binding profiles. A familial binding motif represents the DNA binding preference of closely related TFs. We used the latest versions of motifs in the CORE collection to scan the most popular genomes to predict the TFBSs. Predicted binding sites are available to users along with the genome browser tracks. A new addition in 2022 was word clouds, which summarize the information associated with the TFs in related abstracts from the PubMed database of biomedical literature [131]. The JASPAR database has a user-friendly website, but it is also accessible via the ~~command line~~, RESTful API, R Bioconductor package, and pyJASPAR for advanced users. We provide several computational tools to accompany the data. In this release, we added an enrichment analysis tool that allows users to study the enrichment of the binding of TFs in a set of genomic regions.

the the My contributions to this paper included experimental and computational TF-DNA interaction data mining from repositories and literature, data processing, manual curation of the motifs and their metadata, and revision of the existing motif metadata. In addition, I worked on the update and development of software components, such as the motif discovery pipeline, to obtain candidate DNA binding motifs. I was involved in the downstream analysis, including computation and revision of matrix clusters. Furthermore, I contributed to updating the JASPAR website content. Finally, my contributions include paper writing and coordination of the international team effort.

4.2 Paper II. The release of the 2024 JASPAR database

Rauluseviciute I*, Riudavets-Puig R*, Blanc-Mathieu R, Castro-Mondragon JA, Ferenc K, Kumar V, Lemma RB, Lucas J, Chèneby J, Baranasic D, Khan A, Fornes O, Gundersen S, Johansen M, Hovig E, Lenhard B, Sandelin A, Wasserman WW, Parcy F, Mathelier A. *JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles*. Nucleic Acids Research. 2024 Jan 5;52(D1):D174-82; doi: 10.1093/nar/gkad1059.

*These authors should be considered joint first authors.

4.2. Paper II. The release of the 2024 JASPAR database

Paper II describes the latest release of the JASPAR database. In 2024, the CORE and UNVALIDATED collections were complemented with 404 and 298 profiles, respectively. We validated and moved 72 profiles to the CORE collection. ~~We expanded the insects and nematodes motif collections by 90% and 140%, respectively.~~ Plant TF classification was extensively revised and is now based on Plant-TFClass [54]. One of the most significant updates in the 2024 release was the application of motif trimming. We trimmed 75% (1,869 out of 2,506) of the JASPAR 2022 collection motifs and all incoming new motifs. We applied an algorithm to remove the low IC positions on the flanks of the motifs **to improve the quality of our collections**. The trimming procedure will become a standard when preparing the motifs for manual curation in future releases.

For this release, we also provide genomic tracks of predicted TFBSs in 8 organisms, with mouse and human tracks available as native UCSC Genome browser tracks. In addition, we provide updated motif clusters for each taxon, word clouds, and familial TF binding motifs. We added a TFBS extraction tool to JASPAR's featured tools collection for easier extraction of TFBSs in a genomic region of interest.

The JASPAR database update process is complex, and we have refined it into three operations, abbreviated as "3D": **D**iscovery of the motifs, **D**ownstream analysis, and **D**eployment of the website. Each process has its software component as a pipeline. The motif discovery operation finds new motifs in the **peak** data and processes them for manual curation. It also processes external motifs found in literature or databases by visualizing the motifs as sequence logos and performing comparisons with the current JASPAR motif collections. The downstream analysis prepares manually curated motifs for the website. In addition, the downstream analysis operation performs all additional analyses to produce the data needed for the website. Finally, the deployment workflow collects all produced data and updates the website. All software components received updates to implement new features and increase the reliability and performance of the entire process of JASPAR update.

My contributions to this paper included experimental and computational TF-DNA interaction data mining from repositories and literature, data processing, manual curation of the motifs and their metadata, and revision of the existing motifs and their metadata. I contributed to the software development of motif discovery and downstream analysis pipelines. The motif discovery pipeline was updated to process external motifs along with the discovered ones. I was responsible for motif trimming implementation, application, and comparative analysis of trimmed and untrimmed motifs. In addition, I participated in the curated motif downstream analysis by computing word clouds and familial binding profiles. I contributed to updating the JASPAR website content. Finally, my contributions include paper writing and coordination of the international team effort.

4.3 Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization

Rauluseviciute I, Launay T, Barzaghi G, Nikumbh S, Lenhard B, Krebs AR, Castro-Mondragon JA, Mathelier A. *Identification of transcription factor co-binding patterns with non-negative matrix factorization*. bioRxiv. 2023:2023-04.

Paper III describes a non-negative matrix factorization-based method to identify co-binding DNA patterns. The COBIND tool identifies DNA patterns in neighborhoods of ~~the~~ sequences referred to as anchors. We applied COBIND to a collection of transcription factor binding sites to automatically identify potential TF co-binding patterns in flanking regions. Interpreting such predicted DNA patterns can give insights into TF co-binding and the more complex gene regulation mechanism it creates.

In COBIND, NMF is applied to one-hot encoded sequences and pinpoints the enriched DNA patterns at fixed instances. We used 5,699 TFBS datasets from the UniBind database for 401 TFs in seven species [125]. As a proof-of-concept, we present a classical example of POU5F1 co-binding with SOX2 or SOX17. ~~For~~ SOX2 and SOX17 TFBS datasets, we were able to recover the DNA pattern that represents the POU5F1 motif with different spacings, corresponding to canonical (SOX2::POU5F1 pair) and compressed (SOX17::POU5F1 pair) motifs. Through motif similarity and protein-protein interaction (PPI) data, we inferred that POU5F1 most likely binds the discovered motif.

We evaluated all discovered co-binding patterns with motif similarity and PPI analysis. 78% (462 out of 591) of the patterns had support in the PPI data. For example, TEAD1 is co-binding with TEAD family members. However, some inferred pairs have not yet been reported in the PPI data or do not have a good motif match. In *A. thaliana* genome, the same co-binding pattern was found for HY5, ABF1, ABF3, ABF4, GBF2, and GBF3 anchors. Motif analysis revealed the similarity with the motif recognized by human NF-Y TFs (NF-YC and NF-YA). PPI data does not currently support these interactions. Another example is a co-binding pattern for CTCF. We found a co-binding pattern with two different spacings in *H. sapiens*, *M. musculus* and *D. rerio* TFBS datasets. The literature reported that another TF does not bind the predicted DNA pattern but CTCF itself. CTCF is a zinc-finger protein, and in a fraction of its binding sites, 9–11 zinc fingers bind to an extended portion of the motif.

We demonstrated that the regions with predicted co-binding patterns are likely functionally relevant as they are more evolutionarily conserved than those with only the anchor TFBSs. Across ~~species and~~ all predictions, we systematically compared the conservation at the anchor locations and predicted co-binding sites. We observed that 23–83% of the datasets from different species with predicted co-binding patterns showed significantly higher conservation at the co-binding motif locations compared to the alternative locations in regions only with the anchor motifs. For anchor sites, 31–67% of the datasets demonstrated a more

4.3. Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization

conserved anchor site in the genomic regions with predicted co-binding patterns. We observed for 23–64% of datasets that the locations of both the anchor and co-binding patterns were more conserved in the co-bound regions. More datasets exhibit increased conservation in the co-bound regions compared to what would be expected by chance.

Furthermore, we investigated regions with and without predicted co-binders by analyzing DHS footprints from matching human cell types. We found that in 85% of co-binding pattern and DHS dataset pairs, DHS footprints were significantly deeper at the locations of the co-binding patterns in regions with predicted co-binding motifs compared to the genomic regions without (expectation by chance was 20%). We observed significantly more profound DHS footprints in anchor locations with predicted co-binding patterns for 78% of the pairs (expectation by chance was 21%). Considering deeper footprints for anchor and co-binding pattern sites together, we observed significance for 73% of the pattern-dataset pairs. ~~Furthermore, we observed no significantly deep footprints at both sites in the random expectation.~~

Finally, we utilized published single-molecule footprinting (SMF) data to probe the co-occupancy of TFs at the predicted co-binding patterns. SMF data from mouse embryonic stem cells (mESCs) confirmed that the predicted co-binding events associated with a set of TFs are likely occurring on the same DNA molecules. We analyzed predictions for 17 anchor TFs and found co-binding patterns in 1,694 regions. For each of these genomic regions and regions with only anchor motifs, we determined the proportions of molecules in five states: accessible, occupied by nucleosomes, occupied at the anchor motif site, only occupied at the co-binding motif site, or occupied at both sites. Overall, we found that a significantly more significant fraction of molecules was co-occupied at both sites in the regions with COBIND-predicted co-binding motifs compared to the genomic regions only with the anchor motifs. When looking at the individual anchors, significance was observed for 8 out of 17 analyzed anchor TFs.

We introduced the computational framework COBIND (https://bitbucket.org/CBGR/cobind_tool/src/main/) for DNA motif pattern discovery *de novo* in the vicinity of predetermined regions or a set of sequences. During extensive analysis, we produced a collection of predicted co-binding patterns in 7 species, which we provide to the community at https://cbgr.bitbucket.io/COBIND_results_page/. DNA patterns discovered in the fixed locations from known TFBSs reveal the possible cooperative events and can be studied to investigate more complex TF-DNA interactions.

My contributions to this paper included developing the method, performing the analysis, and preparing the COBIND tool and resource with the results. I analyzed and interpreted the results and wrote the paper.